

Autonomous Ocean Drones for Coral Monitoring

Abstract: The CoralEye network monitored reef health across 42 Pacific locations. Drones identified bleaching events 17 days earlier than traditional surveys.

Introduction: Coral degradation monitoring remains expensive and labor intensive. Autonomous systems may improve coverage.

Methods: Each drone used a hybrid solar-wave power system capable of operating for up to 11 months continuously.

Results: ReefNet-V5 achieved 94.8% image-classification accuracy. Early bleaching detection improved intervention planning.

Discussion: Persistent monitoring enables rapid environmental response and long-term reef trend analysis.

Conclusion: Fictional results demonstrate the value of autonomous marine sensing networks.

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